

OBSERVING THE BEHAVIOUR OF EXISTING STRUCTURES

A Thesis Proposal

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CHAPTER ONE

INTRODUCTION

Background to the Study

Every building must fulfil the safety and usability criteria for its intended purpose throughout the duration of its lifetime of use, according to the general rule of design. In reality, however, any building has a probability of collapsing, regardless of how many safety precautions and variables are included into its design and construction. In addition, many civil infrastructure structures lose part of their strength and rigidity over time due to exposure to severe operational or climatic conditions over their service life (Wang et al., 2017). Multiple structures have collapsed due to environmental conditions and abrupt incidents. This has caused extensive damage and cost the building owners a great deal of money.

Nowadays, structural engineering researchers rely heavily on finite element analysis. This technology may also be used to examine existing concrete bridges that are very large and intricate. This is due to the fact that the finite element model may be updated depending on measurements, and the validated model can be used to examine structural characteristics that cannot be discovered by testing. Experimenting with structures is the greatest approach to determine their true functionality and dependability. Specializes in bridge and traffic structure sensors, as well as the evaluation of selected structural components and structural joints (Dudak et al., 2017; Parkasiewicz et al., 2017; Valluzzi et al., 2018).

Various types of analysis are often required to aid in monitoring and testing. These activities are performed to gain knowledge, but also for technical, economic, or capacity-related objectives. Testing the structural joints and anchors built of precast concrete is crucial. Testing is often conducted on pre-cast concrete panel joints or full-scale beam-to-column junctions (Dal Lago et al., 2018). Experiments often examine the size of the anchoring, the width of the concrete members, the position of the attachment point in the concrete member, the impact of embedment depth on the structure, and the total load on the structure. Numerical studies may be used to augment test programs and provide information that is difficult to quantify or

obtain from experiments by using the finite element approach (Nzabonimpa et al., 2017; Kataoka et al., 2017).

According to Chendo and Obi (2015), a building could collapse due to human errors such as poor design, poor construction, or the use of low-quality building materials, as well as due to negligence, omissions, ignorance, quackery, corruption, or sabotage, or due to natural occurrences such as floods, earthquakes, etc. All of these factors have made it crucial to examine the structural integrity of old buildings to determine if they can continue to function. Wang and Jiang (2015) said that it is essential to analyse the qualities of concrete in order to identify altered regions, manage the quality of the concrete, and determine its compressive strength.

Several existing structures were constructed according to the safety requirements of their day. Due to this, the safety ratings of these structures vary depending on when they were constructed. Therefore, continuing to utilize existing buildings for the same purposes without considering the consequences of wear and tear that have occurred over the course of their service life might be challenging or, in the worst scenario, devastating. Accidents may increase if the condition of older structures is not brought up to current requirements and safety standards (Fabera and Stewart, 2003; Drukis et al., 2017).

Additionally, utilizing ancient buildings in novel ways has become crucial for the preservation of historic structures. Consequently, this entails altering the way an existing system operates from one method to another. Existing concrete structures must frequently be evaluated to see if they can be used today or in the future (Wang and Jiang, 2015). Consequently, it is crucial to conduct a thorough structural examination of existing structures due to the difficulties that these demands provide in the current context (Ma et al., 2012; Drukis et al., 2017; Wang et al., 2017).

Problem Statement

One of the greatest concerns in both emerging and industrialized nations is the constant deterioration of civil engineering buildings (John et al., 2019). Concrete constructions deteriorate too rapidly, resulting in the construction of structures that do not function and are not as safe, sturdy, or aesthetically pleasing as they might be (Alexander, 2018). Consequently, it is often necessary to analyse the many ways in

which actual buildings function (Norhasri et al., 2021). Throughout the years, conventional testing procedures have been employed to determine the functionality of concrete buildings. These techniques, however, do not permit the reuse of test specimens or provide information on the inside of concrete buildings. They are also insufficient to forecast how constructions would behave under adverse conditions, such as when exposed to fire or toxins (Norhasri et al., 2021; Tsioulou et al., 2017). As a result, people are shifting away from destructive procedures and toward non-destructive approaches when evaluating the performance of concrete.

Loreto et al. (2018) conducted a research to determine how a reinforced concrete bridge performs in a maritime environment. This evaluation consisted of three parts: a visual examination of the buildings to determine which portions were impacted by concrete degradation, on-site testing to determine the severity of the deterioration, and laboratory testing to learn more about the deteriorated portions of the structures. The issue with this research is that it employs coring, a destructive testing approach that might compromise the integrity of the building. In addition, the research solely included corrosion as a potential cause of concrete deterioration. This reduced the number of criteria that may be examined to determine the overall safety of the construction.

In a separate research, Aisharqawi et al. (2017) investigated the typical evaluation methods for reinforced concrete buildings. Even if new methods for evaluating the performance of these buildings were recommended, the research mainly discussed visual examination as a method for evaluating reinforced concrete structures, which can only detect external defects and deficiencies. Bossio et al. (2018) investigated the rusting of reinforcing bars in reinforced concrete buildings in a separate investigation.

The research discusses how reinforcing bar corrosion might undermine a building's structure. It also described how High Performance Concrete may be used in areas where corrosion has occurred to make buildings more resistant to collapse; this evaluation is restricted since it provides no further information, such as how much corrosion is acceptable or how much concrete breaks. The thesis is produced so that a comprehensive examination of the strength of structural members made of reinforced concrete may be conducted. This inspection will include the outside and

inside of the members. This thesis will also compare this evaluation's predictions with those of computer software.

Objectives of the Study

The purpose of this research is to determine how existing constructions, in this instance a "T-beam," function by using various measurement techniques, published experimental data, and computer models. The purpose of the research is to compare what various codes say about the behaviour of a reinforced concrete structural element (such as a T-beam) with what the study itself concludes.

1. Using an alternate assessment approach (the compressive force path method)
2. Developing a database of T-beams that have been tested under monotonic static loading to date
3. Using the student edition of ABAQUS to undertake a numerical analysis.

Justification of the Study

The degradation of concrete buildings a few years after construction, which is contrary to the intended design life of concrete structures, is one of the greatest difficulties facing the construction industry. This is due to the use of low-quality materials, poor craftsmanship, and inability to follow to design criteria, amongst other factors (Balamuralikrishnan et al., 2017).

Due to the absence of building integrity assessments, infrastructures degrade more rapidly. To evaluate the performance of existing buildings, it is essential to include non-destructive testing techniques into the quality control process in order to inspect reinforced T-Beams for durability, strength, and fractures using Finite Element Analysis. These approaches not only provide information on the interior status of a building, but they also do so without causing damage to the structure, so enabling the structure to serve its intended purpose.

Scope of Study

This research examines the evaluation of the load-carrying capacity, location, and modes of failure of reinforced concrete structural components by comparing the

predictions of conventional codes (British Standard and Eurocode) with an alternative assessment approach (compressive force path method). It also included comparing the predictions provided by the codes with published experimental data and the outcomes of computer software's numerical projections. This study also provides an overview of previous research and an assessment of the long-lasting reinforced concrete buildings that were predicted following extensive research.

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CHAPTER TWO

LITERATURE REVIEW

Reinforced Concrete Structures

The ability to construct with concrete has existed for a very long period. Concrete is the most popular material used to construct structures such as buildings, bridges, dams, and foundations, etc (Krishna et al., 2016). This is because constructing with concrete is inexpensive, durable, adaptable, practical, and readily available (Fapohunda, 2019; Gupta, 2018; Norhasri et al., 2021). Concrete is a durable construction material produced by combining cement, fine and coarse aggregates, and water in the proper proportions.

In terms of load resistance, unreinforced concrete is strong when compressed but very weak when stretched. Tensile strength accounts for just 10% of concrete's compressive strength (Rodrigues, 2014). Consequently, ordinary concrete can support only a limited amount of weight (Fapohunda, 2019). When loads or shrinkage produce tensile tension, plain concrete fractures. When fractures occur in a simple concrete construction, it collapses entirely and swiftly. The usage of reinforced concrete was initiated to address the shortcomings of plain concrete.

A reinforced concrete structure is a composite construction composed of concrete and steel reinforcing bars. It derives its strength from reinforcing bars placed in places where tensile stresses are exerted and from concrete in regions where compression forces are applied (Aylie et al., 2015; Hammarstrom, 2019). When constructed correctly, reinforced concrete buildings can withstand large loads (concrete carries the compressive and shear loads, while reinforcing steel carries the tensile loads, distributes cracks, and limits the width of the cracks). Obviously, this makes the building more durable (Krishna et al., 2016; Slowik, 2018; Tantawi, 2015).

Due to the link between the reinforcing steel and the concrete in a reinforced concrete structure, this is conceivable. This connection allows compression and tension forces to be transferred from steel to concrete (Rodrigues, 2014). In further detail, the rigidity of a reinforced concrete structure is determined by the concrete between fractures. Because concrete and reinforcing steel join together, this is the

case (Gil Martin et al., 2015). According to Olanitori et al. (2016), the strength of every reinforced structural component is dependent on how it interfaces with other structural elements.

According to them, the most significant aspect of this boundary condition is how it modifies the transmission of internal forces between structural elements. Due to the great compressive strength of concrete and the high tensile strength of reinforcing steel, reinforced concrete is often used for a variety of structural engineering applications, including arches, beams, bridge decks, columns, foundations, piles, shells, slabs, staircases, etc (Fapohunda, 2019; Venkatesh and Alapati, 2017). However, if the building is excessively heavy or not constructed according to the blueprints, it might partly or entirely collapse (Nwaigwe et al., 2019).

Failure in Reinforced Concrete Structures

A major concern is the pace at which civil engineering structures degrade and fall apart (John et al., 2019). This deterioration results in constructions that cannot be used due to their poor appearance, durability, safety, and stability. (Alexander, 2018). Loreto et al. (2018) said that the deterioration of reinforced concrete buildings in coastal locations was due to corrosion induced by chloride attack. According to the research, concrete deterioration manifests as cracks, spalling of the concrete cover, loss of reinforcement in the concrete's cross-section, and loss of bonding between the steel and concrete components of a reinforced concrete structure. Because of this, the structure's capacity to support weight is diminished, and the structure may eventually disintegrate.

Structural failure is the inability of a structure to perform its intended function within the parameters defined by standard codes. In other words, reinforced concrete structures are said to have failed when they can no longer be relied upon to perform their primary functions (Nwaigwe et al., 2019). When a structure is overloaded, its stability and strength may be compromised, leading to structural collapse. (Jamaluddin et al., 2016). Alogla et al. (2016) investigated a phenomenon known as "progressive collapse," which is another means through which a reinforced concrete building might fail. This kind of collapse occurs when there is insufficient continuous

reinforcement and a limited portion of the structure collapses simultaneously with the structural parts becoming less stiff.

The collapse of adjacent portions of a building before the whole structure collapses. Fractures in the reinforcing bars at the point where the beam met the column, crushing of the concrete in the compression zone during flexure, and broad cracks at crucial places where the concrete and reinforcing bars moved significantly indicated the failure of the samples utilized in the research.

Structures in civil engineering may fail for a variety of causes. These include wear and tear, environmental effects, mistakes in design calculations, overuse, poor design and workmanship, incorrect reinforcement spacing, incorrect assignment of loading condition, incorrect type and size of aggregate, incorrect procedure work, lack of integrity checks on existing buildings, load exceeding design, not building according to the contractor's drawing and specifications, inadequate and inconsiderate lap length in structural members, unsuitable materials, and poor (Alsharqawi et al., 2017; Chatterjee et al., 2017; Erhimona and Andrew, 2019; Ganasan et al., 2020; Krishna et al., 2016; Nwaigwe et al., 2019; Rucka, 2020). According to Jamaluddin et al. (2016), a reinforced concrete structure crumbled due to poor construction and weak structural elements.

In reinforced concrete structures, there are typically three forms of failure: bond splitting failure, flexural failure, and shear failure. Bond splitting failure is a kind of bond failure that often occurs when individuals are not constrained. Bond splitting, according to Alabduljabbar et al. (2020), occurs when reinforcing bars are linked by lap splicing. This occurs when fragments of concrete break off between the ribs of the reinforcing bars, causing the link between the concrete and reinforcing bars to fail. According to the research, the bond strength between concrete and reinforcing bars is dependent on anchoring length, bar diameter, concrete cover, presence of transverse reinforcements, concrete strength, etc.

According to a separate research by Bossio et al. (2018), corrosion and a reduction in the cross-section of the reinforcing bars degrade the connection between the concrete and the reinforcing bars. This weakens the structure's integrity. Slowik (2018) investigated why the reinforced concrete structure failed and

concluded that shear failure in buildings is caused by diagonal fractures in reinforced concrete structures that lack transverse reinforcement. According to Dogan and Arslan (2016), shear failure in reinforced concrete structures occurs when the structure's tensile stress exceeds the tensile strength of the concrete.

Rapid shear failure has been identified as the leading cause of reinforced concrete structure failure (Olanitori et al., 2016). Columns are an excellent illustration of the flexural failure mechanism in reinforced concrete buildings, according to Dogan and Arslan (2016). According to them, the concrete fails when it achieves its deformation limit (due to axial loading and bending moment) before the longitudinal reinforcing bars in the tension zone fail.

Lu et al. (2019) examined how the corrosion of the stirrup led to the failure of the reinforced concrete beam. They divided the specimens into three categories: shear failure with stirrup, flexural failure, and shear-compression failure. According to them, shear-compression failure in corroded reinforced concrete is the result of a low stirrup corrosion loss ratio (less than 20%) and a high concrete strength. Conversely, corroded reinforced concrete collapses by shear because to the low concrete strength and high stirrup corrosion loss ratio.

Limit State Design of Reinforced Concrete Structures

It is dangerous when a structure breaks or collapses, or when it does not fulfil the serviceability criteria for cracking, corrosion, deflection, durability, fatigue, fire resistance, and specific situations such as earthquake resistance, vibrations, etc (Fapohunda, 2019). It indicates that a building meets all safety and usability standards for the duration of its design life.

Prior to the introduction of Limit State Design in 1970, "Allowable Stress Design" was used to examine and determine how buildings bend and shear. The Limit State Design was developed to address the restrictions of the Allowable Stress Design, which prohibits load transfer between members (Mohan et al., 2018). Mohan et al. (2018) refer to limit states as instances in which a structure is considered to have "failed to meet its objectives." In other words, limit states are the circumstances or states that a structure must attain prior to being unusable (Fapohunda, 2019).

Mohan et al. (2018) devised three limit states for design: fatigue limit state, serviceability limit state (unfactored loads are utilized to establish if the structure is adequate), and ultimate limit state (the maximum load capacity of a structure is determined using factored loads). But according to Fapohunda (2019), the serviceability limit state of cracking, the condition of deflection, and the ultimate limit state are all required for design calculations in reinforced concrete buildings. The serviceability limit states illustrate the behaviour of the structural part when exposed to the working load. The ultimate limit state, on the other hand, describes the behaviour of the member after it has broken due to bending, compression, shear, or tension (Fapohunda, 2019).

Compressive Force Path Method

The concept behind the compressive force path approach is that the capacity of a reinforced concrete element to withstand shear relies on the concrete's strength in the region that includes the compressive route strength takes as it travels to the supports. Shear failure of a member is influenced by the area's tensile stress build-up (Ayub and Khan, 2017). In other words, according to the compressive force path technique, tensile loads caused shear failure at locations where the shear span-to-depth ratio of a structural element was high (Ahmad et al., 2018).

In a research conducted by Ayub and Khan (2017), the behaviour of four beams was examined using the compressive force path approach and compared to the ACI code method. They discovered that the compressive force path approach was superior to the ACI code method for measuring the shear resistance of reinforced concrete buildings. In addition, the approach may properly forecast theoretical loads; nevertheless, for shear and serviceability, beams must have a shear span-to-depth ratio of less than 4.44. Ahmad et al. (2018) extended to this argument by stating that the compressive force path technique might provide predictions near to the load-bearing capacity, causes, and failures of reinforced concrete structures. In another research by Ahmad et al. (2018), shear failure in reinforced concrete beams or columns occurs when one of the following three circumstances is considered.

1. When the shear span-to-depth ratio, a_v/d , is less than 1. Shear failure occurs when the shear force exceeds $V_{IV} = M_f/a_v$, where M_f is the moment capacity of the beam's cross-flexural section.
2. The ratio of shear span to depth is more than one but less than two and a half. Shear failure occurs when the cross-section at a distance equal to the shear span, a_v , from the point of contraflexure or the nearest support has a bending moment that exceeds $M_{III} = M_{II} + (M_f - M_{II})(2.5 - a_v)/(1.5d)$, where M_{II} is the bending that develops along the length of the reinforced concrete beam at failure.
3. Shear failure occurs when the shear force at a point of contraflexure or 2.5d distant from the support reaches $V_{II, 1} = 0.5ftbd$, where b is the beam width and d is the effective depth.

CHAPTER THREE

METHODOLOGY

T-Beam

T-beams may support a substantial amount of weight due to their resistance or built-in reinforcements. In some aspects, the T-beam predates the first bridge that humans constructed, which consisted of a foundation and a platform. It resembles a column with a flat bed at the top or bottom. A flange is the flat portion of the beam that supports the compression load. A stem is the upright, tension-supporting portion of a plant. Although the materials have evolved through time, the structure has remained unchanged.

However, there are issues with the T-Beam. The T-beam is defective because it lacks a bottom flange. In addition, since the weaker side lacks a flange, the beam has a lower tensile strength, rendering it less usable. Because of this, T-shaped concrete beams are often put into the slab to unite it. The slab portion of these beams supports compressive stresses, while the reinforcing bars at the bottom support tension loads. Typically, a T-stem beam is narrower than a rectangular beam. Therefore, ancient structures with T-beams must be evaluated using numerical analysis to determine their safety.

Numerical Analysis

By creating a mesh, finite element analysis (FEA) disassembles a structure into tens of thousands or even millions of smaller components. Thus, a three-dimensional object is converted into a mathematically comprehensible grid of points. Mathematical equations are employed to predict the behaviour of each component. These computations are then combined to form the structure's final form. Numerous applications of FEA include structural analysis, heat transfer, mass movement, and electromagnetic potential. When designing or constructing anything, you should consider its behaviour in a variety of conditions, both those that can be anticipated and those that cannot.

Even natural calamities such as earthquakes and tsunamis are considered when determining if something may break, wear out, or be damaged in some manner. The unit load and moment distribution approach and the strain energy formula are often used to predict the behaviour of basic constructions (supported beams, trusses). For non-standard objects, you must conduct a comprehensive analysis using structural codes. Rather than depending on costly laboratory studies to determine how a building should respond during construction when there are earthquakes or severe winds, these standards for more accurate and less costly testing.

Earthquakes and other natural catastrophes pose significant challenges to structural engineering, in part because many structures were not constructed according to code and in part because it is difficult to forecast what would occur in these types of calamities. In this research, we will utilize the Abaqus FEA software to execute FEA simulations on three distinct types of T-beams.

Research Beam

Below are the several beams that will be evaluated for the load. Each beam utilizes stirrups.

1. 100 mm x 150 mm x 1000 mm with 2Y10 B/S and T/S. Y8 links @ 100 mm c/c.
2. 100 mm x 150 mm x 1000 mm with 2Y12 B/S and T/S, Y8 links @ 100 mm c/c.
3. 100 mm x 150 mm x 1000 mm with 2Y8 B/S and T/S. Y8 links @ 100 mm c/c.

Loading

You can differentiate between several types of loads. Dead load, Live load, and Impact load are the various forms of design loads that operate on previously constructed buildings. The wind load, longitudinal forces, erection stresses, centrifugal forces, buoyancy effect, deformation effect, and temperature effect all contribute to a structure's performance. However, not every load will be considered for this study. This analysis takes into consideration the dead load, the live load, the

impact load, and the wind load. The loads will be simulated using Abaqus. The structure will be loaded until fractures occur. This will demonstrate the stress tolerance of the structure.

Analysis

The many metrics measured include bending moment, shear force, and strain.

Modelling

During the modelling phase, initial data cleansing occurs. At this point, the input file is created, which instructs the computer on the kind of analysis to do and the solution to use. This research used an iterative linear equation solution. You may use the iterative solver in to discover the solutions to a linear system of equations. It may begin with a static, quasi-static, geostatic, pore fluid diffusion, or heat transfer study. Since the approach is iterative, convergence cannot be guaranteed for a given system of linear equations. However, it is one of the most efficient systems due to the possibility of modifying the model to increase its convergence.

In this phase, the beams are designed, which include determining their dimensions, materials, and other specifics. At this point, the outlines are divided so that the density may be altered, and then the T-beam mesh is created. In addition to the boundary criteria, the T-material Beam's qualities are also predetermined. Also planned are the loading conditions and the number of components.

Processing

Based on the model's solver, we must process the data to do the real analysis. The design is utilized to derive the FEA equations, which are subsequently solved at various phases. The stiffness matrix and the forces at each of the T-Beam element's nodes are computed. Combining and solving the equations yields a nice linear solution. This solution generates output files that display the findings visually.

Reporting

Finally, the findings are presented in an understandable fashion. The outcomes are shown as images, animations, charts, and reports. The findings are shown, and

the properties of the T-output Beam, such as bending moment, shear force, and strain, are determined.

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